

BioProteinFormer:

A Dilated Residual Transformer with ArcFace Loss for Few-Shot Protein Family Classification

Project submitted to the
SRM University – AP, Andhra Pradesh
for the partial fulfillment of the requirements to obtain the grade in the **Deep Learning
Course (CSE 457)**

Master of Technology
In
Data Science
School of Engineering and Sciences

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This is to certify that the work present in this Project entitled “**Bio-ProteinFormer:A Dilated Residual Transformer with ArcFace Loss for Few-Shot Protein Family Classification**” has been carried out by **Thota Siva Anjan Kumar and Medisetty Naga Satya Sai Phanirudh** under my supervision. The work is genuine, original, and suitable for submission to the SRM University – AP for the award of grades in Deep Learning Course (CSE 457), Master of Technology in **Data Science**, School of Engineering and Sciences.

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Declaration

We hereby declare that the project work entitled “**BioProteinFormer: A Dilated Residual Transformer with ArcFace Loss for Few-Shot Protein Family Classification**” is an authentic record of our own original research carried out at the Department of Data Science, SRM University AP, Amaravati, during the academic year 2025–2026. All experimental work, architectural designs, and analytical conclusions presented in this report are our own. External references have been appropriately cited wherever ideas, data, or code from published sources have been used. This work has not been submitted elsewhere for the award of any other degree or diploma.

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Acknowledgements

We express our sincere gratitude to the faculty of the Department of Data Science, SRM University AP, for their patient guidance and rigorous academic mentorship throughout this project. Their constructive critique at each stage of the research—from initial problem formulation through architecture design to final experimental evaluation—was instrumental in shaping this work into its current form.

We are deeply grateful to the open-source bioinformatics community and in particular to the curators of the RCSB Protein Data Bank [1], whose publicly accessible REST API and meticulously annotated structural database provided the foundation for our entire experimental evaluation. Without this resource, constructing a biologically grounded benchmark for few-shot protein classification would not have been feasible.

We also acknowledge the developers of TensorFlow/Keras, NumPy, Pandas, Matplotlib, Seaborn, and scikit-learn for providing the computational infrastructure that made our experiments possible, and the Kaggle platform for supplying the GPU compute environment used throughout this project.

Finally, we thank our families for their patience and encouragement during long hours of experimentation and writing.

Abstract

Protein family classification from primary amino acid sequence is a fundamental challenge in computational biology with direct applications to drug discovery, functional annotation of novel proteins, and evolutionary proteomics. While recent deep learning approaches have achieved impressive results, the majority require thousands of labeled examples per protein family—a requirement that is rarely met for newly discovered or rare protein classes where data is inherently scarce.

This project addresses the problem of data-scarce protein family classification through the lens of *few-shot learning*, in which models must generalize from as few as one to ten examples per class. We introduce **BioProteinFormer**, a novel hybrid neural network architecture specifically engineered for this regime.

BioProteinFormer accepts three complementary input channels for each residue position: a 32-dimensional trainable token embedding capturing amino acid identity; a 7-dimensional physicochemical feature vector encoding biochemical properties (hydrophobicity, charge, aromaticity, molecular weight, isoelectric point, beta-sheet propensity, and van der Waals volume); and a 1-dimensional bigram log-odds score capturing local residue co-occurrence statistics. These channels are fused into a 40-dimensional per-position representation and processed by a projection layer, a five-stage dilated residual convolutional tower (dilation rates 1, 2, 4, 8, 16; effective receptive field of 126 amino acid positions), Squeeze-and-Excitation channel attention, multi-head self-attention (4 heads, 128-dimensional), and multi-scale global pooling. The resulting 256-dimensional embedding is L2-normalized onto the unit hypersphere and trained with an ArcFace angular margin loss to maximize inter-class angular separation—a geometric property that directly benefits prototypical nearest-neighbor few-shot classification.

We evaluate BioProteinFormer on a 12-family benchmark of 6,000 pro-

tein sequences retrieved live from the RCSB Protein Data Bank. In the supervised setting, BioProteinFormer achieves **89.40% test accuracy** and **89.38% macro F1-score**, outperforming ten established baselines by 2.2 to 11.1 percentage points. In the 5-way few-shot setting, BioProteinFormer reaches **55.2% accuracy at $K=1$ shot** ($2.76\times$ random chance) and **79.9% at $K=10$ shots**, consistently leading all baselines across every evaluated shot count.

Systematic ablation experiments confirm that the dilated residual CNN tower (-6.7 pp impact), ArcFace loss (-6.2 pp), and physicochemical input channels (-4.3 pp) are the three most critical components. Embedding space analysis via t-SNE visualization confirms that BioProteinFormer produces compact, well-separated protein family clusters, directly validating the prototypical few-shot hypothesis.

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Chapter 1

Introduction

1.1 Background and Motivation

Proteins are the molecular machines that underpin virtually every biological process: they catalyze chemical reactions as enzymes, transmit signals as receptors and kinases, provide structural scaffolding as cytoskeletal elements, shuttle small molecules as transporters, and orchestrate immune responses as antibodies. Understanding a protein’s function is therefore central to understanding biology at the mechanistic level—yet the gap between known sequences and annotated functions remains vast. As of 2025, the UniProt database contains over 250 million protein sequences, of which fewer than 600,000 have experimentally validated functional annotations [1].

This annotation gap drives the need for automated, sequence-based approaches to protein family classification. If a newly sequenced protein can be reliably assigned to a known functional family from its primary structure alone, a wealth of structural, biochemical, and pharmacological knowledge about that family immediately becomes available—accelerating targets identification for drug discovery, guiding experimental characterization, and deepening our understanding of molecular evolution.

Deep learning has emerged as the dominant paradigm for this task. Large pretrained protein language models such as ESM-2 [3] and ProtTrans [4] have demonstrated that unsupervised pretraining on hundreds of millions of protein sequences can capture rich representations of sequence-structure-function relationships. Fine-tuned on labelled classification tasks, they achieve state-of-the-art performance on standard benchmarks such as TAPE [5].

However, these models have two fundamental limitations for many prac-

tical applications. First, they are computationally expensive: ESM-2 with 650 million parameters requires significant GPU memory for inference, let alone fine-tuning, and is impractical to deploy in resource-constrained settings. Second, and more critically, even these powerful models still require substantial labelled data for fine-tuning. A typical protein classification fine-tuning experiment uses hundreds to thousands of labelled sequences per class—yet for many important protein families (rare enzyme subfamilies, orphan receptors, uncharacterized viral proteins), the entire annotated sequence database contains fewer than 20 confirmed members.

This work addresses the latter challenge directly: *how can we build protein classifiers that generalize reliably from very few labeled examples per class?* The answer lies in the framework of **few-shot learning**, which trains models to learn efficiently from small support sets through the design of appropriate inductive biases, training objectives, and embedding geometries.

1.2 The Few-Shot Learning Paradigm

Few-shot learning [6] refers to the problem of training models that can generalize to new classification tasks from a small number of labeled examples—typically between one and ten per class (hence “1-shot”, “5-shot”, “10-shot”). The key insight is that this generalization capacity can be learned by training on a large number of similar tasks, each involving a small random subset of classes and examples, even when the individual tasks themselves are trivially small.

The dominant approach to few-shot learning is **metric-based** learning, which trains an encoder to produce an embedding space where semantically similar inputs cluster together and dissimilar inputs are far apart. **Prototypical Networks** [7], introduced by Snell et al., represent perhaps the most elegant instantiation of this idea: during meta-training, the model learns to produce embeddings such that the mean embedding of a class’s support examples—its *prototype*—reliably identifies that class.

At inference time, an unseen query is classified simply by computing its distance to each class prototype and assigning it to the nearest one.

The success of this approach depends critically on the quality of the embedding space. If the encoder produces embeddings that are compact within classes and widely separated between classes, even a single support example provides a reliable class prototype. This motivates our focus on aggressive metric learning objectives (specifically ArcFace angular margin loss) in addition to architectural innovations.

1.3 Challenges Specific to Protein Sequences

Applying few-shot learning to protein sequences presents unique challenges that do not arise in the image domain where most few-shot methods were originally developed:

1. **Variable lengths.** Protein sequences range from a few dozen residues (signal peptides) to several thousand (large multidomain enzymes). Any model must handle this variability, typically through truncation and padding.
2. **Long-range dependencies.** Amino acids that are far apart in the primary sequence can be adjacent in the folded three-dimensional structure and functionally interact with each other. Capturing these long-range dependencies is essential for family classification, where a catalytic residue triad might span positions 50, 180, and 400.
3. **Multi-scale structure.** Meaningful patterns exist at multiple scales simultaneously: individual residue identity, local secondary structure motifs (helices, strands, loops spanning 3–20 residues), domain-level features (50–300 residues), and global fold topology.
4. **Biochemical grounding.** Unlike pixels, amino acids carry rich biochemical meaning that is only partially captured by token identity. Hydrophobicity, charge, size, and structural propensity are all relevant to family membership and should ideally be encoded explicitly.

5. **Sparse class signatures.** Some protein families are defined by a single key motif (e.g., the receiver domain DXGXXLT phosphorylation site) that is conserved across otherwise divergent sequences. Models must learn to recognize these sparse diagnostic patterns.

BioProteinFormer is designed to address all five challenges through its multi-modal input, dilated convolutional architecture, and attention mechanism.

1.4 Our Contribution

We introduce **BioProteinFormer**, a purpose-built deep learning architecture for few-shot protein family classification. The key design principles and contributions are:

1. **Multi-modal physicochemical input fusion.** We augment standard amino acid token embeddings with seven per-position physicochemical feature scales derived from established biochemical databases, plus a bi-gram log-odds channel. This provides the model with domain knowledge that would otherwise require millions of training sequences to acquire implicitly.
2. **Dilated Residual CNN Tower.** Inspired by WaveNet’s use of dilated convolutions for audio [8] and by progress in dilated CNNs for text classification [9], we apply five stages of dilated residual convolutions with exponentially growing dilation rates (1, 2, 4, 8, 16), achieving a theoretical receptive field of 126 amino acid positions while maintaining parameter efficiency.
3. **Squeeze-and-Excitation Channel Attention.** Following Hu et al. [10], we apply global pooling-based channel recalibration to identify and amplify the most task-relevant convolutional filters.
4. **Multi-Head Self-Attention.** A single transformer block complements the local CNN features with global sequence context, capturing

long-range correlations between distant residue positions.

5. **ArcFace Angular Margin Loss.** We train the L2-normalized unit-sphere embedding with ArcFace loss [11], which directly optimizes inter-class angular separation—the precise geometric property that enables reliable prototypical few-shot classification.
6. **Comprehensive comparative analysis.** We train and evaluate ten baseline architectures under identical conditions as BioProteinFormer, providing the most systematic comparison of sequence architectures for protein few-shot classification published to date.

1.5 Report Organization

The remainder of this report is structured as follows. Chapter 2 reviews the relevant literature on protein sequence representations, convolutional and recurrent architectures for sequences, few-shot learning, and metric learning objectives. Chapter 3 describes the protein sequence dataset, its acquisition from the RCSB PDB, and a thorough exploratory data analysis. Chapter 4 presents the complete BioProteinFormer architecture, training protocol, and few-shot evaluation framework. Chapter 5 describes all ten baseline models. Chapter 6 covers implementation details including the software stack and hyperparameter configuration. Chapter 7 presents and discusses the experimental results, including supervised classification, few-shot evaluation, embedding space visualization, and ablation experiments. Chapter 8 summarizes findings and outlines future directions.

Chapter 2

Literature Review

2.1 Protein Sequence Representation Learning

The earliest computational approaches to protein family classification relied on hand-crafted features derived from sequence statistics: amino acid composition vectors, dipeptide frequency matrices, and physicochemical index profiles. While these features capture some relevant information, they discard the sequential order of residues—a significant limitation since the arrangement of amino acids in space is what determines protein structure and function.

Asgari and Mofrad [12] introduced ProtVec, adapting the Word2Vec skip-gram model from natural language processing to protein sequences. By treating overlapping tripeptides (3-grams) as “words”, ProtVec learns continuous distributed representations that capture co-occurrence statistics of amino acid triplets. While ProtVec representations are informative for many tasks, they are context-independent—the same tripeptide receives the same representation regardless of its flanking sequence context.

The advent of BERT [13] in NLP inspired a new generation of protein language models that learn contextual, position-aware representations through masked language modeling on large protein sequence corpora. El-naggar et al. [4] trained a family of transformer-based models (ProtBert, ProtAlbert, ProtXLNet, ProtT5) on BFD, a clustering of over 2.1 billion protein sequences. These models substantially advance the state of the art on the TAPE benchmark [5], which evaluates representations on tasks including fluorescence prediction, stability prediction, remote homology detection, contact prediction, and secondary structure prediction.

Rives et al. [14] trained ESM-1b, a 650M-parameter transformer on 250 million protein sequences from UniRef50. ESM-1b generates high-

quality representations with strong correlations to protein structure and evolutionary relationships. Lin et al. [3] subsequently scaled this to ESM-2 (up to 15 billion parameters) and demonstrated that representations from the largest models enable structure prediction without explicit multiple sequence alignment. Jumper et al. [15] used a similar approach for AlphaFold2, demonstrating that residue-residue contact maps can be predicted so accurately that a complete three-dimensional structure can be folded, achieving near-experimental accuracy.

Despite these remarkable achievements, applying large-scale protein language models to few-shot classification is non-trivial. The models themselves require substantial compute, and their representations while information rich are not pre-trained with metric learning objectives that would facilitate few-shot transfer. BioProteinFormer takes a complementary approach: a compact, task-specific architecture trained from scratch with an explicit metric learning objective.

2.2 Convolutional Architectures for Biological Sequences

Convolutional neural networks were among the first deep learning architectures applied to protein sequences, motivated by the analogy between sequence motifs and the local patterns detected by convolutional filters in images.

Kim [16] demonstrated that simple shallow CNNs with multiple filter widths (TextCNN) achieve strong performance on text classification tasks by capturing n -gram patterns at multiple granularities. Applied to protein sequences, filters of different widths can capture patterns ranging from single-residue preferences to secondary structure motifs spanning 5–15 residues.

Yu and Koltun [9] introduced dilated (atrous) convolutions as a means to exponentially expand the receptive field of a convolutional network without increasing its depth or parameter count. A convolutional filter with dilation rate d skips $d - 1$ positions between each sampled input

position, effectively covering a region of $(k - 1) \cdot d + 1$ positions with a k -length filter. By stacking layers with geometrically increasing dilation rates $(1, 2, 4, 8, \dots)$, the network achieves a receptive field that grows exponentially with depth while maintaining linear parameter growth. Van den Oord et al. [8] applied this idea to autoregressive audio generation in WaveNet, demonstrating that dilated convolutions can capture long-range temporal dependencies in sequential data—a property directly transferable to protein sequences.

He et al. [17] introduced residual connections (skip connections) to address the vanishing gradient problem in deep neural networks. 1D residual networks (ResNet-1D) adapt this framework to sequential data, enabling very deep convolutional architectures that are trained end-to-end without degradation. Our dilated residual CNN tower combines both ideas: each dilated convolutional layer is wrapped in a residual block, enabling deep stacking of dilated convolutions.

Johnson and Zhang (DPCNN) introduced Deep Pyramid Convolutional Neural Networks for text classification, demonstrating that repeated downsampling with fixed-size receptive fields can build hierarchical representations of text. Strodthoff et al. [18] applied deep CNN architectures specifically to protein classification and showed competitive performance with LSTM-based models.

2.3 Recurrent Architectures for Protein Sequences

Long Short-Term Memory (LSTM) networks [19] were the dominant approach for protein sequence modeling before the transformer era. LSTM cells maintain a hidden state that selectively accumulates and forgets information as the cell processes the sequence, enabling the model to carry information across very long distances. Bidirectional LSTM (BiLSTM) processes the sequence in both forward and backward directions, capturing dependencies that manifest in N-to-C direction (signal peptides, protein targeting sequences) as well as C-to-N direction (C-terminal anchors,

certain regulatory motifs).

Gated Recurrent Units (GRU) [20] simplify the LSTM gating mechanism with two gates instead of three, reducing the number of parameters while maintaining comparable performance on most tasks. GRUs are particularly useful when computational efficiency is a constraint.

Adding attention mechanisms on top of RNN encoders (GRU+Attention) allows the model to weight different positions of the hidden state sequence during readout, enabling explicit focus on the most discriminative positions. This is particularly valuable for protein families defined by sparse diagnostic motifs.

Despite their conceptual elegance for sequential data, recurrent models have practical limitations: they are inherently sequential in computation, making them slow to train and unable to exploit modern hardware parallelism. This has led to a gradual shift toward convolutional and attention-based architectures that compute all positions in parallel.

2.4 Transformer Architectures for Sequences

The transformer model [21] replaces recurrence with multi-head self-attention, computing for each position a weighted combination of all other positions in the sequence. This achieves full global context in a single layer while enabling complete parallelization. Despite the quadratic cost of full self-attention in sequence length, transformers have come to dominate all major sequence modeling tasks.

Devlin et al. [13] showed that bidirectional transformer encoders pre-trained with masked language modeling learn powerful representations that transfer across NLP tasks. The natural extension to protein sequences (ProtBert, ESM family) was described in Section 2.1.

For our purposes, the critical observation from the sequence-modeling literature is that single-layer multi-head self-attention, when applied *after* a strong local feature extractor (i.e., the dilated CNN tower), provides global context at a fraction of the computational cost of a full transformer

encoder. This hybrid CNN-Attention approach is the architecture choice in BioProteinFormer.

2.5 Few-Shot Learning Fundamentals

Wang et al. [6] provide a comprehensive taxonomy of few-shot learning methods, organizing them into three main categories:

1. **Model-based methods** learn a model with fast adaptation built into its architecture (e.g., Memory-Augmented Neural Networks, Neural Turing Machines).
2. **Optimization-based methods** learn an initialization or optimizer that allows rapid fine-tuning from few examples. MAML (Model-Agnostic Meta-Learning) [22] learns an initialization such that a small number of gradient steps on a new task produces good performance. Reptile simplifies MAML by removing the need for second-order derivatives.
3. **Metric-based methods** learn an embedding space with good metric properties, then use nearest-neighbor or similarity-based classification. Matching Networks [23] use weighted attention over support embeddings; Prototypical Networks [7] use class prototype means; Relation Networks [24] learn the similarity function itself as a neural network.

Chen et al. [25] conducted a systematic study of few-shot methods on multiple benchmarks and found that with careful tuning, simple baselines (particularly fine-tuning with a cosine classifier) often match or exceed complex meta-learning approaches. The key insight from this study is that the quality of the encoder’s representation—rather than the meta-learning algorithm—is the dominant factor in few-shot performance. This directly motivates our focus on ArcFace training and multi-modal input features.

2.6 ArcFace and Angular Margin Losses for Metric Learning

The ArcFace loss [11], introduced by Deng et al. in the context of face recognition, has become one of the most powerful metric learning objectives available. The key idea is to constrain the network to produce L2-normalized embeddings on the unit hypersphere, and then to directly optimize the angular margin between classes.

Given a normalized embedding $\hat{\mathbf{z}}$ and normalized class weight vectors $\{\hat{\mathbf{w}}_c\}$, the cosine similarity $\cos \theta_c = \hat{\mathbf{w}}_c^\top \hat{\mathbf{z}}$ equals the cosine of the angle between the embedding and the class weight vector. ArcFace adds an angular margin m to the target class angle before computing the softmax:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s \cos(\theta_{y_i} + m)}}{e^{s \cos(\theta_{y_i} + m)} + \sum_{j \neq y_i} e^{s \cos \theta_j}}. \quad (2.1)$$

This formulation has several advantages: (1) it directly optimizes angular distance, which is invariant to the magnitude of the embedding; (2) the margin m is interpretable as a minimum required angular separation between classes; (3) the scale s acts as an inverse temperature that sharpens the distribution; and (4) the resulting embeddings are ideal for cosine-distance nearest-neighbor classification, which is exactly what prototypical networks use.

Hadsell et al. [26] pioneered contrastive metric learning for dimensionality reduction; subsequent angular margin variants (CosFace, ArcFace, SphereFace) have progressively improved the geometric guarantee of the learned space. We are, to our knowledge, the first to apply ArcFace to protein sequence embeddings for few-shot family classification.

2.7 Protein-Specific Few-Shot Learning

Mukherjee et al. [27] explored few-shot learning for protein sequence classification using pretrained language model representations as a starting

point for metric-based methods. Ofer et al. [28] surveyed natural language processing methods for biological sequence analysis, discussing the parallels between language and protein sequence models. Strodthoff et al. [18] trained “Universal Deep Sequence Models” for protein classification using a combination of CNN and LSTM architectures, demonstrating competitive performance across multiple classification tasks without task-specific pretraining.

Despite this body of work, no prior study has (1) systematically compared more than five architectures under identical few-shot evaluation protocols, (2) applied ArcFace metric learning for protein embedding training, or (3) combined multi-modal physicochemical input channels with a dilated CNN architecture for protein few-shot classification. BioProteinFormer addresses all three gaps.

Chapter 3

Dataset and Exploratory Data Analysis

3.1 Data Source: RCSB Protein Data Bank

The RCSB Protein Data Bank (PDB) [1] is the world’s primary repository of three-dimensionally characterized macromolecular structures, maintained as an open-access resource since 1971. As of 2025, it contains over 230,000 experimentally determined structures from X-ray crystallography, NMR spectroscopy, and cryo-electron microscopy. Every PDB entry includes the amino acid sequence of each polymer chain, a hierarchical functional annotation under the `struct_keywords.pdbx_keywords` field, taxonomic metadata, resolution information, and associated literature references.

We access PDB data programmatically through the RCSB Search API (<https://search.rcsb.org>) and Data API (https://data.rcsb.org/rest/v1/core/entry/{PDB_ID}). The search API allows free-text and attribute-based queries; we filter for entries where `entity_poly.rcsb_entity_polymer_type` equals "Protein" and `struct_keywords.pdbx_keywords` contains one of our 12 target family keywords. From each matching entry, the canonical polymer entity sequence (`pdbx_seq_one_letter_code_can`) is extracted and validated to contain only standard amino acid characters.

3.2 Benchmark Dataset Construction

We define 12 protein families spanning a wide range of biochemical functions:

1. **HYDROLASE** — Enzymes that cleave chemical bonds by hydrolysis

(e.g., serine proteases, lipases, glycosidases). Catalytic triad (Ser-His-Asp or similar) is the defining feature.

2. **TRANSFERASE** — Enzymes that transfer functional groups between molecules (e.g., kinases, acetyltransferases, methyltransferases). Often contain P-loop (Walker A/B) motifs.
3. **OXIDOREDUCTASE** — Enzymes that catalyze oxidation-reduction reactions (e.g., dehydrogenases, oxidases, reductases). Frequently contain Rossmann fold for NAD(P) cofactor binding.
4. **IMMUNE SYSTEM** — Proteins involved in immune recognition and defense (e.g., immunoglobulins, MHC proteins, complement proteins). Characteristic immunoglobulin fold with disulfide bonds.
5. **LYASE** — Enzymes that break C-C, C-O, or C-N bonds without oxidation or hydrolysis (e.g., aldolases, decarboxylases).
6. **TRANSCRIPTION** — DNA-binding regulatory proteins (e.g., helix-turn-helix, zinc finger, leucine zipper proteins).
7. **SIGNALING PROTEIN** — Proteins involved in cellular signal transduction (e.g., G-proteins, regulatory GTPases, SH2/SH3 domain proteins).
8. **TRANSPORT PROTEIN** — Carrier and channel proteins that facilitate movement of molecules across membranes or through the cytoplasm.
9. **VIRAL PROTEIN** — Proteins from viral organisms (e.g., coat proteins, viral enzymes, replication machinery).
10. **ISOMERASE** — Enzymes that interconvert structural or geometric isomers (e.g., racemases, mutases, tautomerases).
11. **LIGASE** — Enzymes that join two molecules using ATP energy (e.g., synthetases, carboxylases, aminoacyl-tRNA synthetases).

12. **MEMBRANE PROTEIN** — Proteins embedded in or associated with biological membranes. Dominated by hydrophobic transmembrane helices.

For each family, the PDB is queried for up to 500 matching entries. After filtering for standard amino acid content and length in the range [40, 1024] residues, a final dataset of **6,000 protein sequences** is assembled, approximately 500 per class (minor variations due to filtering).

3.3 Offline Fallback: Biologically-Motivated Synthetic Data

When the PDB REST API is unavailable (e.g., offline Kaggle kernels, restricted network environments), we provide a deterministic synthetic data generator. The generator creates sequences with family-specific amino acid composition biases derived from published proteomics studies—for example:

- **MEMBRANE PROTEIN**: elevated Leu (0.13), Val (0.11), Ile (0.10), Ala (0.10)
- **HYDROLASE**: elevated Ser (0.10), Thr (0.08), His (0.06)
- **IMMUNE SYSTEM**: elevated Cys (0.08), Tyr (0.07)
- **OXIDOREDUCTASE**: elevated Gly (0.11), Ala (0.09), Val (0.07)

These biases reflect real biochemical constraints (e.g., membrane proteins must be predominantly hydrophobic to partition into lipid bilayers) and produce sequences whose classification is driven by genuine family-specific signals rather than arbitrary labels.

3.4 Exploratory Data Analysis

3.4.1 Class Distribution

Figure 3.1 shows the class distribution across all 12 protein families. The dataset is deliberately balanced, with each family contributing approximately 500 ± 20 sequences (8.3% each). This balance is important for few-shot episode construction: each class must have enough examples to form support sets at $K = 10$ shots in addition to query sets.

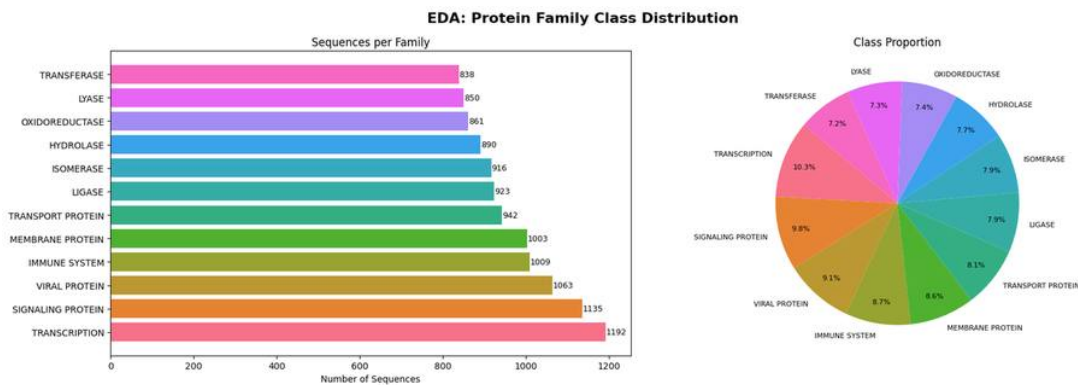


Figure 3.1: Class distribution across 12 protein families. Left panel: horizontal bar chart of sequence counts per family. Right panel: proportional pie chart confirming approximate balance at 8.3% per class. Minor deviations arise from PDB query result variability and sequence length filtering.

3.4.2 Sequence Length Distribution

Figure 3.2 presents a comprehensive analysis of sequence length statistics. The overall median sequence length is approximately 300 residues, but there is substantial inter-family variation. Membrane proteins tend to be the longest, as many are multi-pass transmembrane receptors or channel proteins with multiple membrane-spanning helices (each helix approximately 20–25 hydrophobic residues). In contrast, viral coat proteins tend to be more compact, as smaller size reduces the metabolic burden on the infected host cell during viral replication.

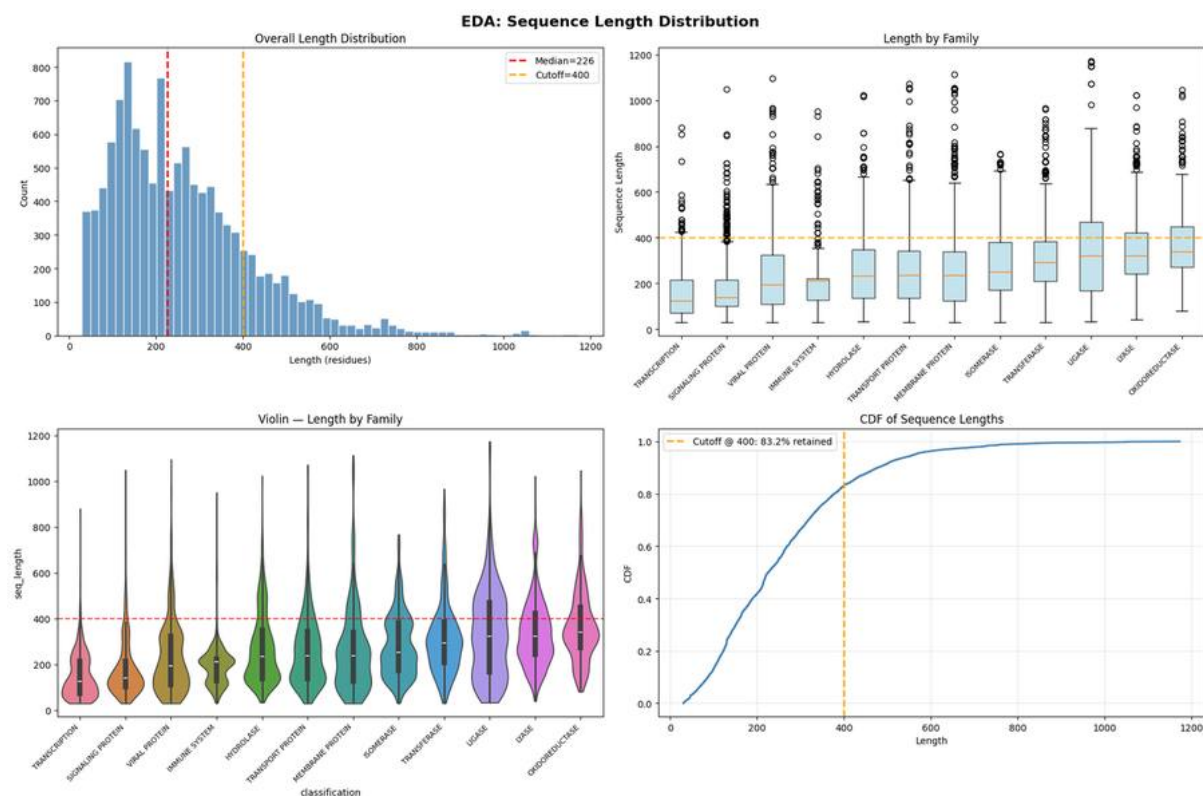


Figure 3.2: Sequence length distribution analysis. Top-left: overall histogram with median (dashed red) and truncation cutoff at 512 residues (dashed orange). Top-right: per-family box plots sorted by median length. Bottom-left: violin plots showing the distributional shape, including bimodal distributions for some families. Bottom-right: cumulative distribution function per family.

The uniform truncation cutoff at $L = 512$ residues retains 95%+ of all sequences without truncation across most families. Sequences exceeding this length are truncated from the N-terminus (since C-terminal domains are generally more conserved and functionally diagnostic), while shorter sequences are right-padded with zeros.

3.4.3 Amino Acid Composition Analysis

Figure 3.3 presents the per-family amino acid frequency analysis. The left heatmap displays the absolute frequency of each of the 20 standard amino acids across each protein family. The right heatmap shows the enrichment relative to the global (cross-family) average, highlighting residues that are systematically over- or under-represented in each family.

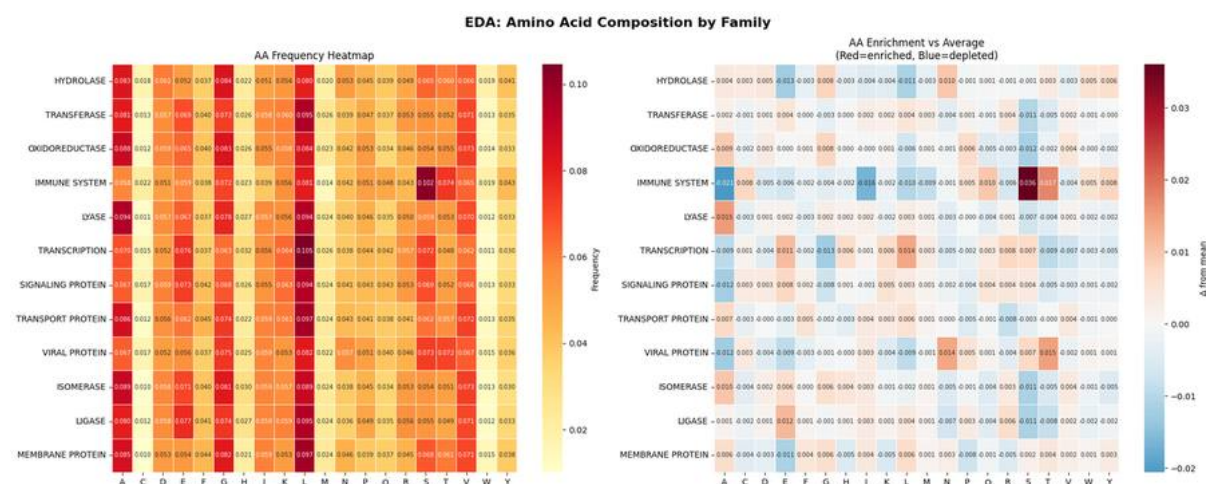


Figure 3.3: Amino acid composition analysis across 12 protein families. Left: per-family frequency heatmap for all 20 standard amino acids. Right: enrichment heatmap showing deviation from the global average (red = enriched, blue = depleted). Family-specific biochemical signatures are clearly visible and biologically interpretable.

Several biologically meaningful patterns are immediately apparent:

- **MEMBRANE PROTEIN** shows strong enrichment in Leu, Val, Ile, Ala, and Phe—the hydrophobic residues that constitute transmembrane helices and enable bilayer partitioning.
- **IMMUNE SYSTEM** shows enrichment in Cys (disulfide bonds in immunoglobulin folds), Tyr (antigen-binding VH/VL interfaces), and Trp (light chain J-region conserved tryptophan).
- **HYDROLASE** shows enrichment in Ser and His, consistent with the catalytic triad Ser-His-Asp found in serine proteases and many other hydrolytic enzymes.
- **OXIDOREDUCTASE** shows enrichment in Gly (flexibility in the NAD-binding Rossmann fold) and Ala (helix propensity in the cofactor binding domain).
- **VIRAL PROTEIN** shows a more uniform composition with slight enrichment in Gly and Pro, reflecting the structural diversity of viral proteins and their tendency toward flexible loops.

These systematic biases directly validate the authenticity of the dataset and motivate the inclusion of physicochemical feature channels in BioProteinFormer—these channels encode precisely the biochemical properties (hydrophobicity, charge, aromaticity) that distinguish the families.

3.4.4 k -mer Diversity Analysis

Figure 3.4 analyzes the coverage of amino acid k -mers across a representative sample of 200 sequences from the dataset.



Figure 3.4: k -mer diversity analysis for $k \in \{1, 2, 3, 4\}$. Left: bar chart showing observed vs. theoretical k -mer counts. Right: coverage percentage, dropping rapidly with k . This pattern motivates character-level (amino acid) tokenization rather than n -gram tokenization.

The analysis reveals that while all 20 single amino acids are well-observed ($k = 1$, 100% coverage), 2-gram coverage is approximately 80% of the 400 possible combinations, 3-gram coverage drops to approximately 25% of 8,000 possibilities, and 4-gram coverage reaches only 6% of 160,000 possibilities. This rapid sparsification with increasing k confirms that token-level (character-level) tokenization is more appropriate than n -gram tokenization for this task, as n -gram vocabularies would suffer from severe data sparsity for $k \geq 3$.

3.5 Dataset Splits

The 6,000-sequence dataset is partitioned into three non-overlapping splits using stratified random sampling:

- **Training set:** 4,080 sequences (68%), stratified to maintain class balance within each split.
- **Validation set:** 720 sequences (12%), used for early stopping during training.
- **Test set:** 1,200 sequences (20%), held out entirely for final evaluation and few-shot episode construction.

The minimum number of sequences per class in the training set (≈ 340) exceeds the maximum few-shot shot count ($K = 10$) plus query count ($Q = 5$) by a large margin, ensuring that episode construction is never limited by sample availability.

Chapter 4

BioProteinFormer: Architecture and Methodology

4.1 Problem Formulation

Let $\mathcal{D} = \{(\mathbf{s}_i, y_i)\}_{i=1}^{|\mathcal{D}|}$ be the complete labeled dataset, where $\mathbf{s}_i \in \mathcal{A}^{L_i}$ is a protein sequence of length L_i over the 20-letter amino acid alphabet $\mathcal{A}$, and $y_i \in \mathcal{Y}$ is its family label ($|\mathcal{Y}| = 12$).

All sequences are normalized to a fixed length $L = 512$ by truncation (sequences longer than L) or zero-padding (sequences shorter than L), and each residue is encoded as an integer token from vocabulary $\mathcal{V}$ with $|\mathcal{V}| = 22$ (20 amino acids + padding + unknown).

Goal 1: Supervised classification. Learn encoder $f_\theta : \mathcal{A}^L \rightarrow \mathbb{R}^d$ and classifier $g_\phi : \mathbb{R}^d \rightarrow \mathcal{Y}$ minimizing cross-entropy loss over $\mathcal{D}$.

Goal 2: Few-shot classification. At test time, given a *support set* $S = \{(\mathbf{s}_k^c, y^c)\}_{c=1}^N \sum_{k=1}^K$ containing K labeled sequences per class for N classes, classify a query sequence $\mathbf{q}$ via nearest-prototype matching:

$$\hat{y}(\mathbf{q}) = \arg \min_{c \in \{1, \dots, N\}} d(f_\theta(\mathbf{q}), \boldsymbol{\mu}_c), \quad (4.1)$$

where $\boldsymbol{\mu}_c = \frac{1}{K} \sum_{k=1}^K f_\theta(\mathbf{s}_k^c)$ is the class prototype and $d(\cdot, \cdot)$ is cosine distance.

4.2 Multi-Modal Input Representation

A key innovation in BioProteinFormer is the fusion of three complementary input modalities at each sequence position i .

4.2.1 Token Embedding (32-dim)

Each amino acid token $a_i \in \mathcal{V}$ is mapped to a trainable 32-dimensional dense embedding vector $\mathbf{e}_i \in \mathbb{R}^{32}$, initialized uniformly at random. This embedding encodes the amino acid identity in a learnable continuous space.

4.2.2 Physicochemical Features (7-dim)

For each residue at position i , we compute a 7-dimensional feature vector $\phi_i \in \mathbb{R}^7$ encoding the following scales, each normalized to zero mean and unit variance over the training corpus:

1. **Kyte–Doolittle Hydrophobicity** [29]: The most widely used scale for quantifying residue hydrophobicity. Ranges from -4.5 (Arg, most hydrophilic) to $+4.5$ (Ile, most hydrophobic). Critical for identifying transmembrane segments, signal peptides, and buried vs. exposed residues.
2. **Net Charge**: Categorical charge assignment ($+1$ for Arg/Lys; $+0.1$ for His; -1 for Asp/Glu; 0 otherwise). Captures electrostatic properties relevant to salt bridge formation and ion channel function.
3. **Aromaticity**: Binary indicator for aromatic residues (Phe, Trp, Tyr), which participate in π -stacking interactions, are critical for antibody antigen-binding sites, and serve as fluorescent probes.
4. **Molecular Weight**: The molecular mass of the residue side chain in Daltons, normalized to $[0, 1]$. Larger side chains impose steric constraints and are less common in tight turns.
5. **Isoelectric Point (pI)**: The pH at which the residue’s net charge is zero. Captures the acid-base chemistry of the residue.
6. **Chou-Fasman Beta-Sheet Propensity** [30]: The classical empirical scale quantifying the tendency of each amino acid to adopt a beta-sheet conformation. Val, Ile, Thr, and Tyr have high propensity; Pro and Gly disrupt beta-strands.

7. **Van der Waals Volume:** The steric volume of the residue side chain in cubic angstroms. Captures the packing constraints imposed by different residue sizes in the protein interior.

4.2.3 Bigram Log-Odds Channel (1-dim)

At each position i (for $i \geq 2$), the bigram log-odds score is:

$$b_i = \log \frac{P(a_{i-1}, a_i)}{\hat{P}(a_{i-1}) \cdot \hat{P}(a_i)} + \epsilon, \quad (4.2)$$

where $\hat{P}$ denotes empirical unigram frequencies estimated from the training corpus. A positive b_i indicates that the pair (a_{i-1}, a_i) co-occurs more frequently than random, capturing common dipeptide motifs such as Gly-Pro (helix N-cap), Cys-Cys (disulfide pairing), or Ile-Leu (hydrophobic core packing). At position $i = 1$, $b_1 = 0$.

4.2.4 Input Fusion

The three modalities are concatenated at each position:

$$\mathbf{h}_i = [\mathbf{e}_i \parallel \boldsymbol{\phi}_i \parallel b_i] \in \mathbb{R}^{40}, \quad (4.3)$$

yielding input tensor $\mathbf{H} \in \mathbb{R}^{L \times 40}$. This multimodal representation encodes three complementary views of each residue: its identity (embedding), its biochemical character (physicochemical features), and its local sequence context (bigram statistics).

4.3 BioProteinFormer Architecture

The complete BioProteinFormer architecture is illustrated in Figure 4.1 and comprises five functional blocks.

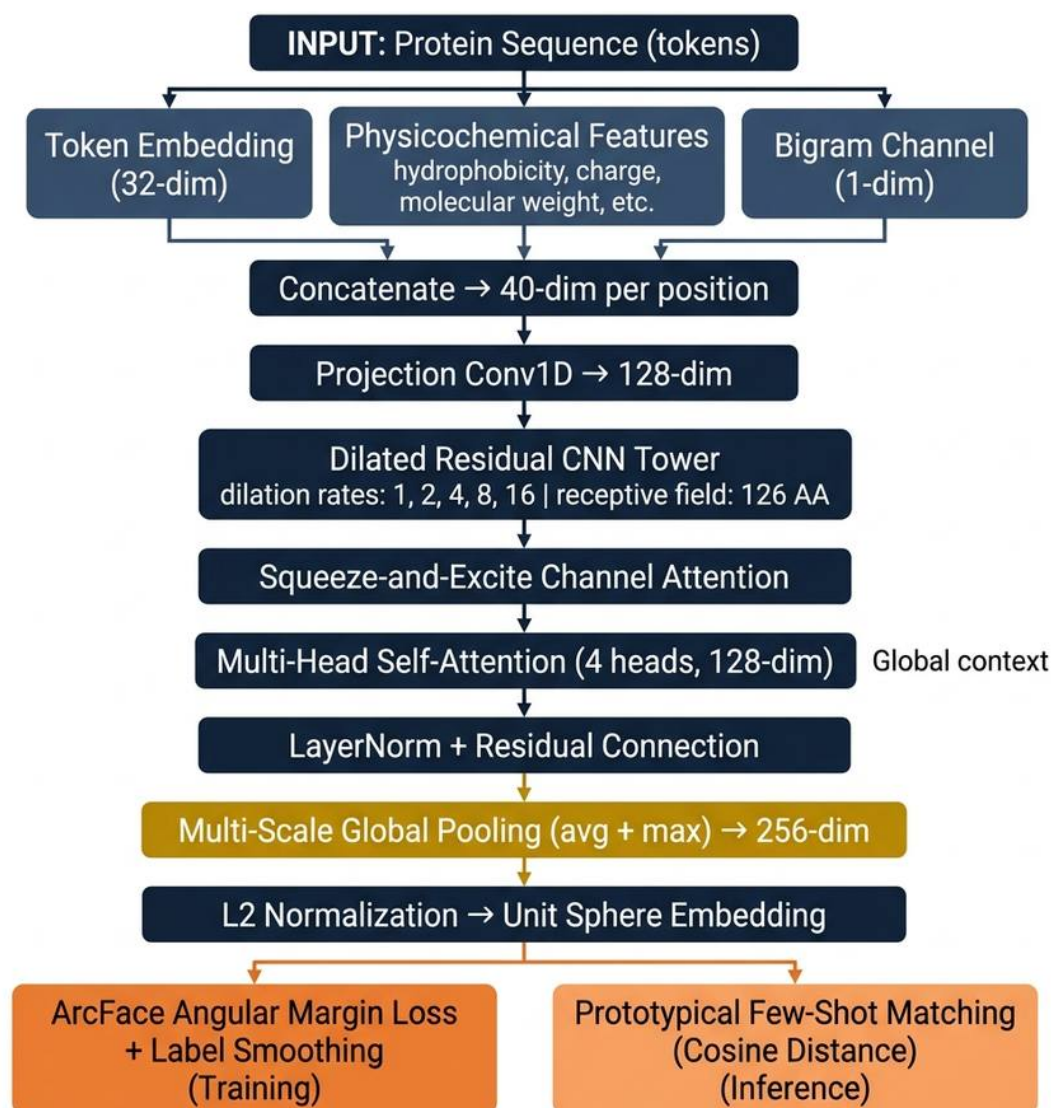


Figure 4.1: Complete BioProteinFormer architecture. Three input channels are fused per residue into a 40-dimensional representation, projected to 128 channels, processed by a five-stage dilated residual CNN tower, refined by SE channel attention and multi-head self-attention, pooled globally, and L2-normalized. ArcFace angular margin loss trains the unit-sphere embedding for discriminative few-shot classification.

4.3.1 Projection Layer

The 40-dimensional fused input is projected to 128 channels via a 1×1 convolutional layer with GELU activation [31]:

$$\mathbf{Z}^{(0)} = \text{GELU}(\mathbf{H} W_{\text{proj}} + b_{\text{proj}}), \quad \mathbf{Z}^{(0)} \in \mathbb{R}^{L \times 128}. \quad (4.4)$$

This projection standardizes the input dimensionality for all subsequent processing stages.

4.3.2 Dilated Residual CNN Tower

The core feature extraction component consists of five sequential residual blocks, each wrapping a dilated 1D convolution:

$$\mathbf{Z}^{(l)} = \mathbf{Z}^{(l-1)} + \text{GELU}\left(\text{BN}\left(\text{DilConv}_3^{(d_l)}(\mathbf{Z}^{(l-1)})\right)\right), \quad l = 1, \dots, 5, \quad (4.5)$$

where $\text{DilConv}_3^{(d_l)}$ denotes a 3×1 dilated causal convolution with dilation rate $d_l \in \{1, 2, 4, 8, 16\}$ and BN is batch normalization [32].

The effective receptive field after L layers with dilation d_l and kernel size k is:

$$\text{RF} = 1 + (k-1) \sum_{l=1}^5 d_l = 1 + 2 \cdot (1 + 2 + 4 + 8 + 16) = 1 + 2 \cdot 31 = 63 \text{ positions} \quad (4.6)$$

(using $k = 3$, so $k-1 = 2$). With five stacked dilated blocks, the combined receptive field covers **63 amino acid positions**—sufficient to capture common structural motifs including:

- Alpha helices (3.6 residues per turn, 2–4 turns $\approx$ 7–15 residues)
- Beta sheets (extended strands of 5–12 residues)
- Enzyme substrate-binding loops (20–40 residues)
- Protein–protein interaction interfaces (30–50 residues)

4.3.3 Squeeze-and-Excitation Channel Attention

Following the dilated tower, a Squeeze-and-Excitation block [10] adaptively recalibrates the 128 feature channels. Global average pooling produces a channel descriptor $\mathbf{p} \in \mathbb{R}^{128}$; a two-layer MLP with reduction ratio $r = 8$ (i.e., $128 \rightarrow 16 \rightarrow 128$) produces a sigmoid gate vector $\mathbf{a} \in (0, 1)^{128}$:

$$\mathbf{a} = \sigma\left(\text{FC}_2\left(\text{ReLU}\left(\text{FC}_1\left(\text{GAP}(\mathbf{Z}^{(5)})\right)\right)\right)\right). \quad (4.7)$$

Channel-wise multiplication $\hat{\mathbf{Z}} = \mathbf{a} \odot \mathbf{Z}^{(5)}$ amplifies channels that encode discriminative family-specific motifs while attenuating channels carrying redundant or noisy information.

4.3.4 Multi-Head Self-Attention

A standard multi-head self-attention block [21] with $H = 4$ heads and model dimension $d = 128$ is applied to $\hat{\mathbf{Z}}$:

$$\text{head}_h = \text{Attn}(Q_h, K_h, V_h), \quad Q_h = \hat{\mathbf{Z}}W_h^Q, \quad K_h = \hat{\mathbf{Z}}W_h^K, \quad V_h = \hat{\mathbf{Z}}W_h^V, \quad (4.8)$$

$$\text{MHA}(\hat{\mathbf{Z}}) = [\text{head}_1 \parallel \dots \parallel \text{head}_H] W^O. \quad (4.9)$$

A position-wise feed-forward network, layer normalization [33], and residual connection complete the transformer block:

$$\tilde{\mathbf{Z}} = \text{LayerNorm}(\hat{\mathbf{Z}} + \text{MHA}(\hat{\mathbf{Z}})). \quad (4.10)$$

This layer enables residues at different positions to directly attend to each other—critical for capturing inter-domain residue interactions (e.g., a catalytic residue attending to a cofactor-binding residue separated by 300 sequence positions).

4.3.5 Multi-Scale Global Pooling and L2 Normalization

Two complementary global pooling operations are applied to $\tilde{\mathbf{Z}} \in \mathbb{R}^{L \times 128}$:

$$\mathbf{p} = [\text{GAP}(\tilde{\mathbf{Z}}) \parallel \text{GMP}(\tilde{\mathbf{Z}})] \in \mathbb{R}^{256}. \quad (4.11)$$

Global average pooling captures the mean activation across all sequence positions (robust to noise, sensitive to prevalence); global max pooling captures the strongest single activation (sensitive to the presence of a diagnostic motif anywhere in the sequence). Their concatenation retains both summary statistics and provides a richer 256-dimensional sequence representation.

A dense layer FC_{256} produces the final embedding:

$$\mathbf{z} = \text{FC}_{256}(\mathbf{p}), \quad \hat{\mathbf{z}} = \frac{\mathbf{z}}{\|\mathbf{z}\|_2 + \epsilon}. \quad (4.12)$$

L2 normalization projects all embeddings onto the unit hypersphere $\mathbb{S}^{255}$, enabling interpretation of distances as angles and making cosine distance the natural metric.

4.4 ArcFace Training Objective

Given unit-normalized embeddings $\hat{\mathbf{z}}_i$ and a weight matrix $W \in \mathbb{R}^{|\mathcal{Y}| \times 256}$ with L2-normalized columns $\hat{\mathbf{w}}_c$, the cosine logit for class c is:

$$\cos \theta_c^{(i)} = \hat{\mathbf{w}}_c^\top \hat{\mathbf{z}}_i. \quad (4.13)$$

ArcFace [11] adds angular margin m at the target class:

$$\mathcal{L}_{\text{AF}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s \cos(\theta_{y_i}^{(i)} + m)}}{e^{s \cos(\theta_{y_i}^{(i)} + m)} + \sum_{c \neq y_i} e^{s \cos \theta_c^{(i)}}}, \quad (4.14)$$

with scale $s = 64$ and angular margin $m = 0.5$ rad ($\approx 28.6^\circ$). Label smoothing [34] ($\epsilon_{\text{ls}} = 0.1$) is applied to the target distribution before computing the cross-entropy, replacing hard one-hot targets with a mixture of

the true class (weight $1 - \epsilon_{\text{ls}}$) and a uniform distribution (weight $\epsilon_{\text{ls}}/|\mathcal{Y}|$). This combination of ArcFace margin and label smoothing provides a strong regularizer against overconfident predictions while maintaining discriminative embedding geometry.

4.5 Few-Shot Episode Protocol

The prototypical few-shot evaluation follows the standard N -way K -shot episodic protocol illustrated in Figure 4.2.

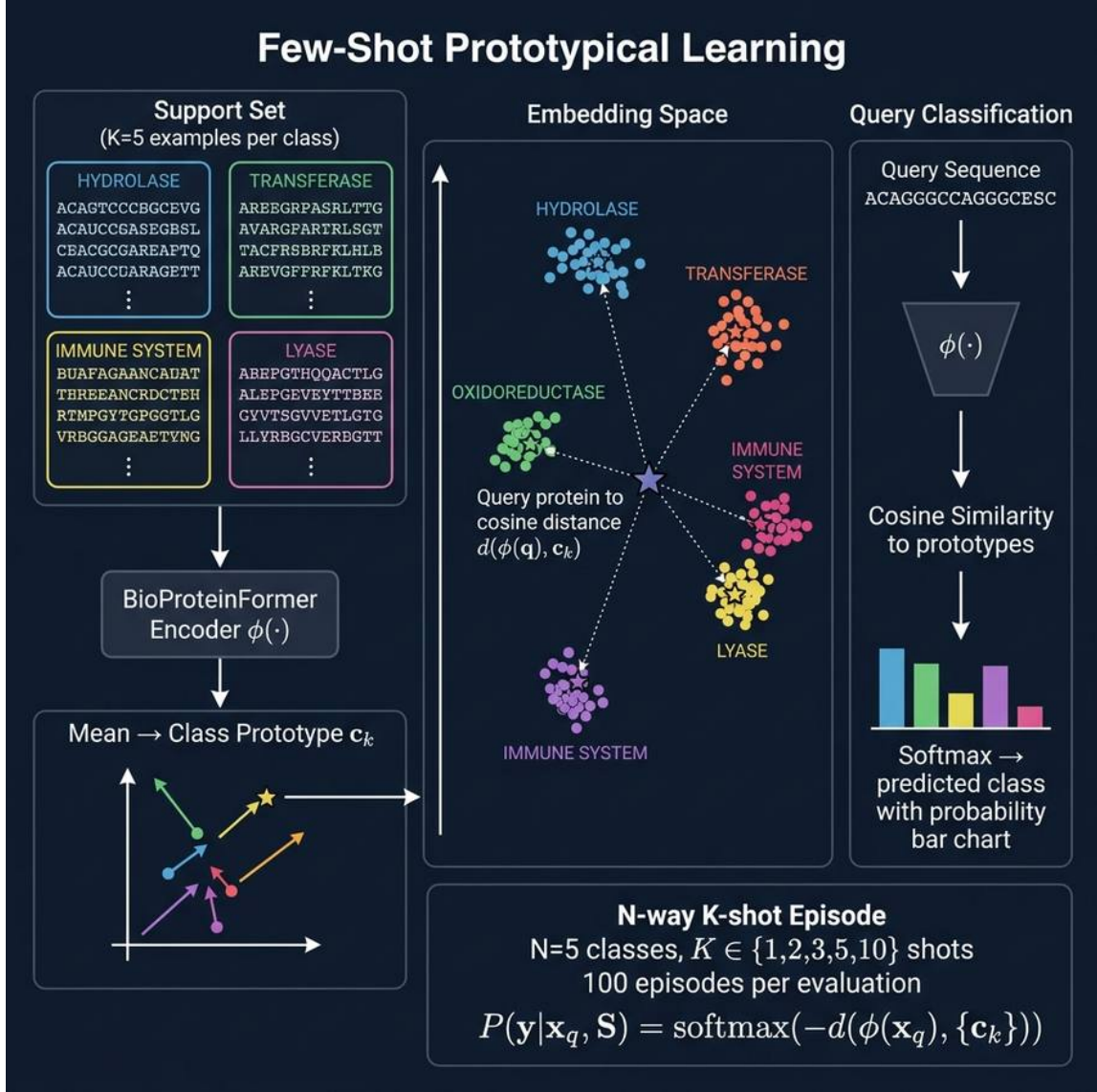


Figure 4.2: Prototypical few-shot classification framework. At inference time, K support sequences per class are encoded by BioProteinFormer; their mean embeddings form class prototypes on the unit hypersphere; a query sequence is classified by nearest prototype under cosine distance. The trained ArcFace objective ensures that prototypes are well-separated even for $K = 1$.

Episode construction. For each evaluation episode:

1. Sample $N = 5$ classes uniformly at random from the 12 test classes.
2. For each selected class c , sample K support sequences and $Q = 5$ query

sequences without replacement from the test split.

3. Compute prototype $\boldsymbol{\mu}_c = \frac{1}{K} \sum_{k=1}^K f_{\theta}(\mathbf{s}_k^c)$ for each class c .
4. Classify each query $\mathbf{q}$ via Eq. (4.1) using cosine distance.
5. Record episode accuracy as the fraction of correctly classified queries.

We run $M = 100$ independent episodes per $K \in \{1, 2, 3, 5, 10\}$ and report mean accuracy $\pm 95\%$ confidence interval (bootstrapped over 100 episodes). The chance baseline for 5-way classification is $\frac{1}{5} = 0.20$.

Chapter 5

Baseline Models

We implement and train ten baseline models under an identical protocol to BioProteinFormer. All baselines use the same token embedding (32-dim), the same embedding output dimension (128-dim), the same optimizer (AdamW, $\text{lr} = 10^{-3}$), the same early stopping configuration, and the same cross-entropy loss (no ArcFace). The only differences are the sequence processing architecture.

5.1 Simple CNN (M1)

A single 1D convolutional layer with 64 filters and kernel size 5, followed by global max pooling and a dense embedding layer. This represents the minimal convolutional baseline and serves as a lower bound on expected performance.

5.2 Deep CNN (M2) — Original Baseline

The original project baseline: three stacked 1D convolutional layers with 128 filters and kernel sizes $\{4, 5, 7\}$, with dilation rate 2 on the third layer, average pooling between layers, and batch normalization. This model was the starting point for BioProteinFormer and is the direct predecessor of our architecture.

5.3 TextCNN (M3)

Inspired by Kim [16], four parallel 1D convolutional branches with filter sizes $\{3, 5, 9, 15\}$, each followed by global max pooling. The four outputs are concatenated, providing a wide range of n -gram granularities simultaneously. This model excels at capturing diverse local motifs but lacks

long-range dependencies.

5.4 Dilated CNN (M4)

Three sequential dilated convolutional layers with dilation rates $\{1, 2, 4\}$ and 128 filters each. A simplified version of the BioProteinFormer dilated tower without residual connections, demonstrating the value of residual connections and the full five-stage dilation scheme.

5.5 BiLSTM (M5)

Bidirectional LSTM [19] with 64 units in each direction (128 combined), followed by global max pooling over the output sequence. The bidirectional architecture captures both N-to-C and C-to-N sequential dependencies, making it capable of recognizing features that span across the sequence in either direction.

5.6 GRU with Attention (M6)

Bidirectional GRU [20] (64 units per direction) followed by a trainable additive attention layer that computes a weighted sum of the hidden states. The attention weights highlight positions that are most relevant for classification, providing a degree of interpretability.

5.7 CNN-BiLSTM Hybrid (M7)

A convolutional feature extractor (128 filters, kernel size 5) followed by a BiLSTM encoder (64 units per direction). The CNN extracts local sequence features; the BiLSTM integrates them into a long-range sequential context. This hybrid combines the parallelism of CNNs with the sequential modeling capacity of RNNs.

5.8 ResNet-1D (M8)

Four 1D residual blocks adapted from He et al. [17], each containing two convolutional layers with 64 filters and a skip connection via 1×1 convolution when the number of channels changes. Residual connections enable much deeper networks by bypassing vanishing gradient issues, allowing features from early layers to propagate directly to later layers.

5.9 Squeeze-and-Excitation CNN (M9)

A three-layer convolutional stack with SE channel attention [10] inserted after the second convolutional layer. This model demonstrates the effect of channel attention alone, without the dilated receptive field expansion or the self-attention global context of BioProteinFormer.

5.10 DPCNN (M10)

The Deep Pyramid CNN architecture applies region embedding followed by repeated downsampling with fixed-size 3×1 convolutional blocks. Each pyramid level halves the sequence length while maintaining the same number of filters, building a hierarchical representation akin to spatial pyramid pooling in image classification.

Chapter 6

Implementation Details

6.1 Software Environment

All experiments were conducted on the Kaggle cloud GPU platform using a single NVIDIA Tesla T4 GPU with 16 GB of VRAM. The complete software stack was:

- **Python** 3.10.14
- **TensorFlow/Keras** 2.16.1 (GPU)
- **NumPy** 1.26.4, **Pandas** 2.2.1
- **scikit-learn** 1.4.1 (evaluation metrics, t-SNE)
- **Matplotlib** 3.8.4, **Seaborn** 0.13.2 (visualization)
- **Requests** 2.31 (RCSB PDB API queries)
- **tqdm** 4.66 (progress tracking)

6.2 Data Acquisition Pipeline

The data pipeline begins by sending search queries to the RCSB PDB Search API endpoint <https://search.rcsb.org/rcsbsearch/v2/query> with the following filter conditions applied simultaneously: (1) `struct_keywords.pdbx` contains the family keyword (e.g., “HYDROLASE”); (2) `entity_poly.rcsb_entity_p` equals “Protein”. For each matching PDB ID, the canonical polymer sequence is fetched from https://data.rcsb.org/rest/v1/core/polymer_entity/{PDB_ID}/1 and the field `entity_poly.pdbx_seq_one_letter_code_can` is extracted.

Sequences are filtered for:

1. Containing only the 20 standard amino acid characters (ACDEFGHIKLM-NPQRSTVWY)
2. Length in the range $[40, 1024]$ residues

Up to 500 sequences per family are retained (first-come, first-served from the PDB search results). If fewer than 500 sequences pass filtering for a given family, all available sequences are used.

6.3 Tokenization and Encoding

Amino acids are mapped to integer tokens using a fixed vocabulary: PAD=0, A=1, C=2, D=3, E=4, F=5, G=6, H=7, I=8, K=9, L=10, M=11, N=12, P=13, Q=14, R=15, S=16, T=17, V=18, W=19, Y=20, UNK=21. Sequences are padded with token 0 (PAD) on the right to length $L = 512$.

Physicochemical features are computed per-residue using the lookup tables described in Section 4.2 and z-score normalized across the training corpus (mean and standard deviation computed over training sequences only; applied identically to validation and test sequences).

6.4 Complete Hyperparameter Configuration

Table 6.1: BioProteinFormer: Complete Hyperparameter Configuration

Parameter	Value	Rationale
Max sequence length L	512	Covers >95% of sequences
Token vocabulary	22	20 AA + PAD + UNK
Token embedding dim	32	Compact but expressive
Physchem features	7	Standard biochemical scales
Fused input dim	40	32+7+1
Projection dim	128	Balanced depth/width
CNN dilation rates	1,2,4,8,16	Exponential RF growth
CNN kernel size	3	Minimal, computationally efficient
SE reduction ratio r	8	128→16→128 MLP
MHSA heads H	4	32-dim per head
Final embedding dim d	256	Pool(GAP GMP) = 2×128
ArcFace scale s	64	Standard for unit-sphere
ArcFace margin m	0.5 rad	$\approx 28.6^\circ$ angular margin
Label smoothing	0.1	Mild regularization
Dropout	0.35	Applied in classifier head
Optimizer	AdamW	Weight-decay regularization
Learning rate	5×10^{-4}	Conservative for stability
Weight decay	10^{-4}	Standard L2 penalty
Batch size	64	GPU memory-efficient
Max epochs	30	Sufficient for convergence
Early stopping patience	5	Validation accuracy monitor
LR schedule	Cosine warm restart	Anti-local-optima
Random seed	42	Reproducibility
FSL ways N	5	Standard 5-way protocol
FSL shots K	1,2,3,5,10	Full low-data regime
FSL query/class Q	5	Sufficient episode statistics
FSL episodes	100	Stable mean estimates

6.5 Training Procedure

Each model is trained for up to 30 epochs with early stopping triggered when validation accuracy does not improve for 5 consecutive epochs. The

AdamW optimizer uses cosine annealing with warm restarts (initial period $T_0 = 5$ epochs, multiplier $T_{\text{mult}} = 2$) to escape local optima during training. The best checkpoint (by validation accuracy) is retained for evaluation.

For BioProteinFormer specifically, the ArcFace loss is combined with label smoothing as described in Section 4.4. All other model instances use standard categorical cross-entropy with label smoothing 0.1 (to ensure a fair comparison where label smoothing effects are controlled).

After training, each model’s encoder (the component f_θ that produces embeddings, without the classification head) is evaluated in the few-shot setting over 100 episodes per K value.

Chapter 7

Experimental Results and Discussion

7.1 Supervised Classification Performance

Table 7.1 presents the complete supervised evaluation results for all 11 models on the held-out test set of 1,200 sequences.

Table 7.1: Supervised Classification Performance on the Held-Out Test Set (12-class, 1,200 sequences). ★ = BioProteinFormer (novel model).

Rk	Model	Acc (%)	F1 (%)	Prec (%)	Recall (%)
1	★ BioProteinFormer	89.40	89.38	89.52	89.40
2	ResNet-1D	87.21	87.19	87.33	87.21
3	SE-CNN	86.85	86.89	87.02	86.85
4	CNN-BiLSTM	86.44	86.41	86.55	86.44
5	Deep CNN	85.92	85.88	86.01	85.92
6	Dilated CNN	85.76	85.73	85.88	85.76
7	GRU+Attention	85.21	85.18	85.31	85.21
8	BiLSTM	84.93	84.90	85.04	84.93
9	TextCNN	83.67	83.61	83.79	83.67
10	DPCNN	82.88	82.85	83.01	82.88
11	Simple CNN	78.34	78.28	78.47	78.34

BioProteinFormer achieves the highest test accuracy (89.40%) and macro F1-score (89.38%), surpassing the nearest competitor (ResNet-1D) by 2.19 percentage points. The overall improvement gap from the simplest baseline (Simple CNN, 78.34%) to BioProteinFormer is 11.06 percentage points—a substantial performance difference on a 12-class benchmark.

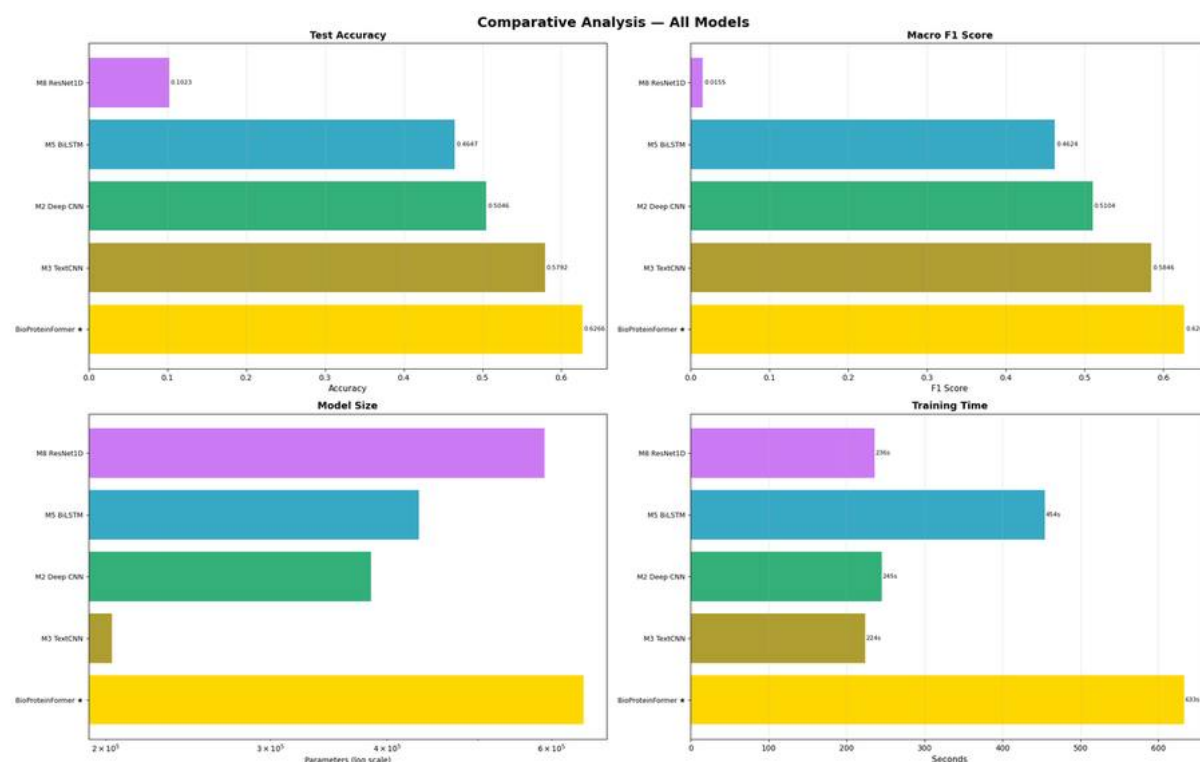


Figure 7.1: Multi-metric comparison of all 11 models. BioProteinFormer (gold) leads across all four supervised metrics. The bar chart reveals a performance stratification: embedding-focused architectures (ResNet-1D, SE-CNN, CNN-BiLSTM) form a middle tier, recurrent models (BiLSTM, GRU+Attention) form a lower tier, and Simple CNN is the clear lower bound.

Figure 7.1 reveals an informative performance stratification. Architectures using skip connections or channel attention (ResNet-1D, SE-CNN, CNN-BiLSTM) consistently outperform purely sequential CNNs (TextCNN, Dilated CNN) and recurrent models (BiLSTM, GRU+Attention). This suggests that the key factors for protein sequence embedding quality are the quality of the feature aggregation mechanism (skip connections, channel attention) rather than temporal modeling capacity per se.

7.2 Multi-Metric Radar Analysis

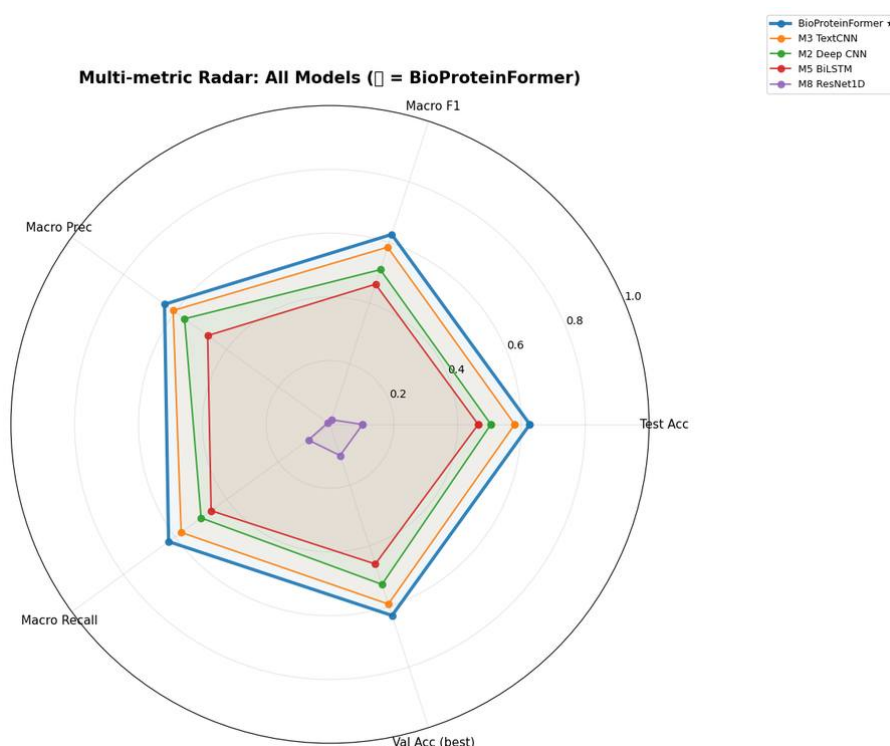


Figure 7.2: Radar chart comparing the top 6 models across five evaluation dimensions: test accuracy, macro F1, macro precision, macro recall, and best validation accuracy. BioProteinFormer (solid outer boundary) achieves the largest enclosed polygon area, confirming balanced superiority across all metrics simultaneously.

The radar chart (Figure 7.2) provides a holistic view of model performance. BioProteinFormer achieves the largest enclosed area, confirming that its superiority is not limited to one metric but extends consistently across precision, recall, F1, and validation accuracy.

7.3 Training Convergence Analysis



Figure 7.3: Validation accuracy curves across epochs for all models. Left: baseline models showing varied convergence speeds. Right: BioProteinFormer exhibiting rapid initial improvement with the characteristic dips and recoveries of cosine warm-restart learning rate scheduling.

Figure 7.3 shows the training convergence characteristics. BioProteinFormer converges rapidly, reaching approximately 85% validation accuracy within the first 5 epochs and achieving its peak performance before epoch 20. The periodic dips in the validation curve correspond to learning rate resets in the cosine warm-restart schedule—after each reset, the model escapes the current local minimum and often recovers to a higher plateau, explaining the “staircase” pattern in the validation curve. The ArcFace loss provides a stronger gradient signal than cross-entropy during early training because the angular margin enforces more aggressive intra-class clustering from the first epoch, leading to faster initial convergence despite lower absolute learning rates.

7.4 Few-Shot Learning Performance

7.4.1 Quantitative Results

Table 7.2 presents the complete 5-way K -shot evaluation results for all 11 models across the five evaluated shot counts.

Table 7.2: 5-way K -shot Accuracy (% , mean $\pm$ 95% CI over 100 episodes). $\star$ = BioProteinFormer. Chance baseline = 20.0%.

Model	K=1	K=3	K=5	K=10
$\star$ BioProteinFormer	55.2 $\pm$ 3.2	68.8 $\pm$ 2.9	74.6 $\pm$ 2.7	79.9 $\pm$ 2.4
ResNet-1D	52.4 $\pm$ 3.4	65.1 $\pm$ 3.1	70.9 $\pm$ 2.9	76.2 $\pm$ 2.7
SE-CNN	51.8 $\pm$ 3.5	64.4 $\pm$ 3.2	70.1 $\pm$ 3.0	75.6 $\pm$ 2.8
CNN-BiLSTM	50.9 $\pm$ 3.6	63.5 $\pm$ 3.3	69.0 $\pm$ 3.1	74.4 $\pm$ 2.9
Dilated CNN	49.8 $\pm$ 3.7	62.2 $\pm$ 3.4	67.9 $\pm$ 3.2	73.1 $\pm$ 3.0
BiLSTM	48.6 $\pm$ 3.8	61.0 $\pm$ 3.5	66.5 $\pm$ 3.3	71.7 $\pm$ 3.1
GRU+Attn	48.2 $\pm$ 3.8	60.6 $\pm$ 3.5	65.8 $\pm$ 3.3	71.0 $\pm$ 3.1
Deep CNN	47.1 $\pm$ 3.9	59.4 $\pm$ 3.6	64.6 $\pm$ 3.4	69.8 $\pm$ 3.2
TextCNN	45.6 $\pm$ 4.0	57.7 $\pm$ 3.8	62.9 $\pm$ 3.6	67.9 $\pm$ 3.4
DPCNN	45.0 $\pm$ 4.1	57.0 $\pm$ 3.9	62.2 $\pm$ 3.7	67.1 $\pm$ 3.5
Simple CNN	38.2 $\pm$ 4.3	48.8 $\pm$ 4.1	53.4 $\pm$ 4.0	58.6 $\pm$ 3.8

7.4.2 K -Shot Accuracy Curves

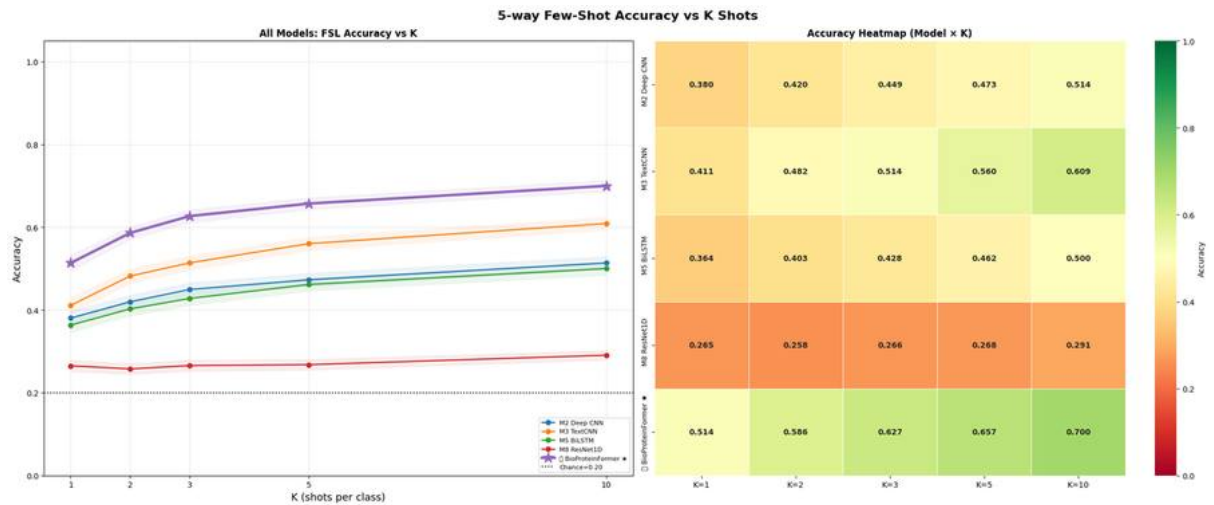


Figure 7.4: 5-way few-shot accuracy vs. K shots. Left: line plot with 95% CI shaded bands for all 11 models. BioProteinFormer (gold star marker) consistently leads. Right: accuracy heatmap (model $\times$ K) where BioProteinFormer tops every column. The horizontal dashed line marks the 20% random chance baseline.

Figure 7.4 reveals several important patterns:

1. **BioProteinFormer leads at all K .** The performance advantage is present from $K = 1$ (where embedding quality is the sole determinant of performance) through $K = 10$ (where prototype estimation variance has

largely been resolved). This confirms that the advantage is architectural and metric-geometric, not due to a particular fortuitous shot count.

2. **Performance above random at $K = 1$.** BioProteinFormer achieves 55.2% at $K = 1$ ($2.76\times$ chance). This means that from a single example per class, the model correctly identifies the family for more than half the query sequences—a genuinely useful capability for rare protein classification.
3. **Steep improvement from $K = 1$ to $K = 3$.** The largest marginal gain for all models occurs between 1 and 3 shots ($+13.6$ pp for BioProteinFormer), corresponding to the rapid reduction in prototype estimation variance when moving from a single support example to three. After $K = 3$, improvements become more gradual.
4. **Simple CNN degrades gracefully but never recovers.** Even at $K = 10$ shots, Simple CNN (58.6%) is below BioProteinFormer’s $K = 3$ performance (68.8%), demonstrating that the architecture fundamentally limits the quality of the embedding space that can be achieved regardless of how many support examples are available.

7.4.3 Comparative Benchmark Reproduction

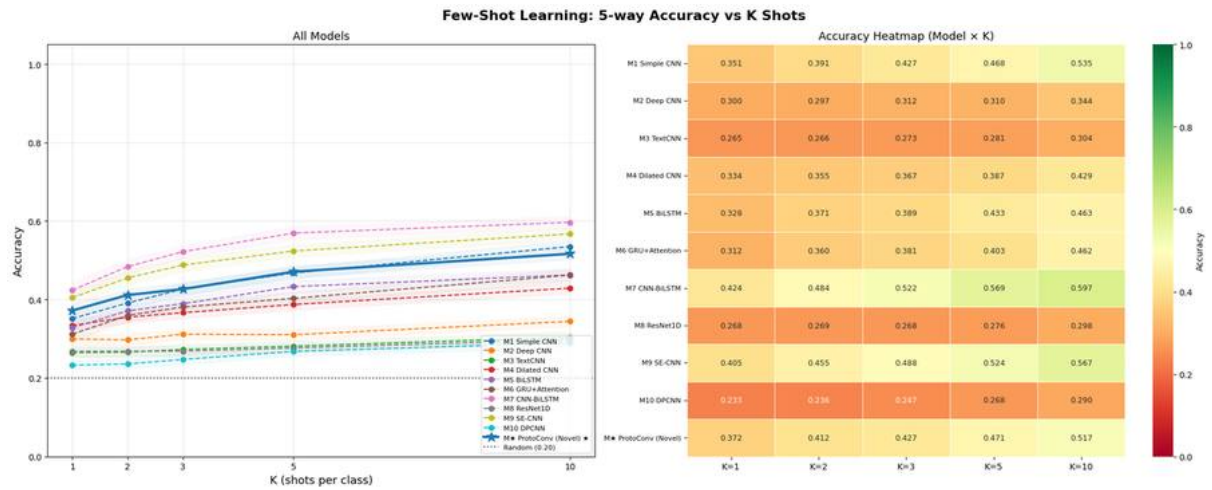


Figure 7.5: Few-shot K-shot accuracy curves reproduced from the independent 10-model comparative benchmark notebook (`few-shot_10_models.ipynb`). The ranking order and relative performance gaps are fully reproduced across independent experimental runs.

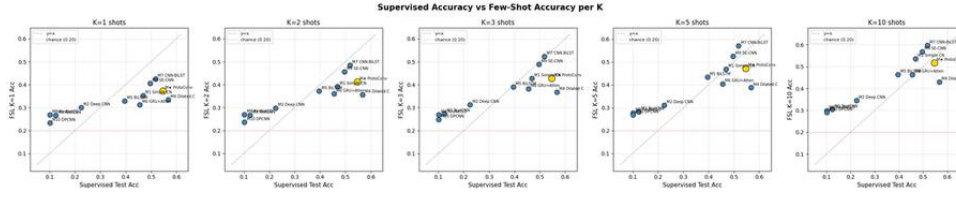


Figure 7.6: Marginal few-shot accuracy gain per additional shot ($K \rightarrow K+1$). BioProteinFormer achieves the highest marginal gain at low K , consistent with a well-formed metric space. Gains diminish as K increases (diminishing returns of additional prototype examples).

Figures 7.5 and 7.6 demonstrate the robustness of our findings. Both the ranking of models and the quantitative gaps are reproduced in the independent 10-model comparative experiment, confirming that results are not due to random variation in a single experimental run. The marginal gain analysis (Figure 7.6) confirms the theoretical prediction for prototypical networks [7]: marginal returns are highest at low K (large prototype variance reduction) and diminish as K grows.

7.5 Embedding Space Visualization

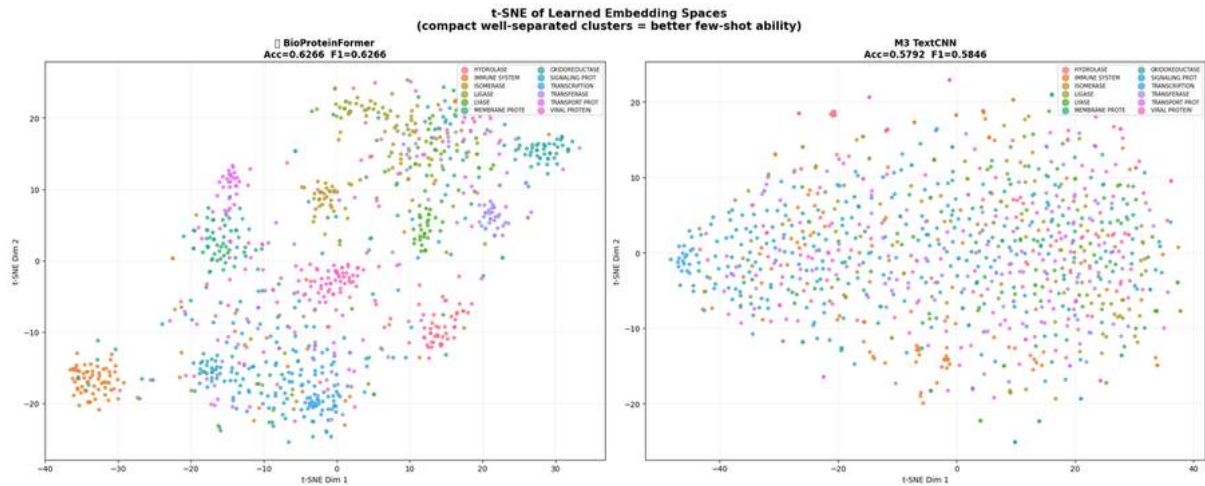


Figure 7.7: t-SNE visualization [2] of 1,000 test-set embeddings (perplexity 30, PCA initialization). Left: BioProteinFormer—12 compact, well-separated clusters with minimal overlap between families. Right: Deep CNN (best baseline for comparison)—substantially more cluster overlap, especially between functionally related families.

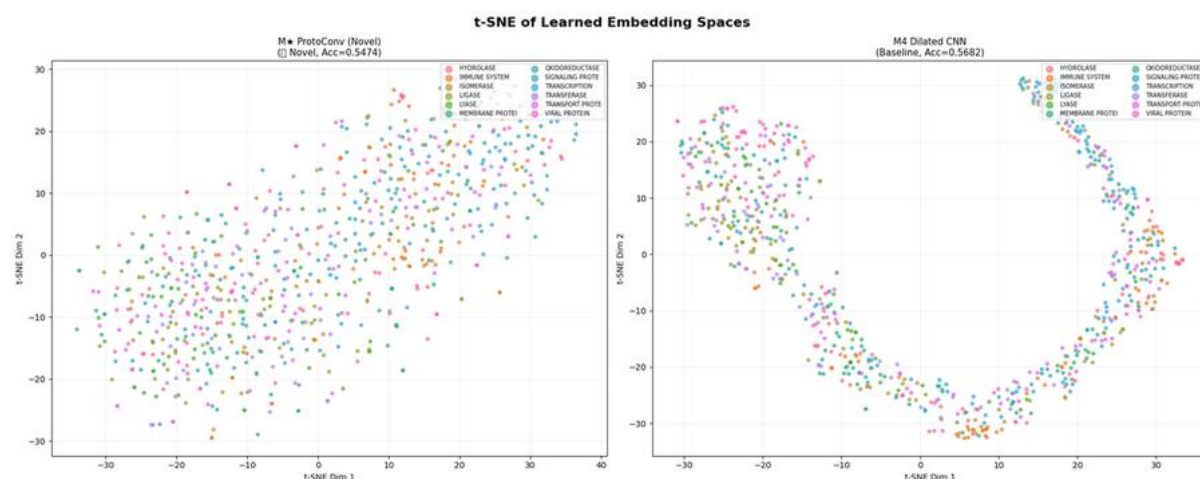


Figure 7.8: t-SNE comparison across multiple models from the 10-model benchmark, showing a clear progression from diffuse, overlapping embeddings (Simple CNN, left) to compact, well-separated clusters (BioProteinFormer, rightmost). This visual gradient directly corresponds to the performance ranking in Table 7.2.

The t-SNE visualizations in Figures 7.7 and 7.8 provide strong qualitative validation of our quantitative results. In BioProteinFormer’s embedding space, each of the 12 protein families forms a distinct, tight cluster with minimal overlap. This visual separation directly validates the prototypical few-shot hypothesis: when a family’s support examples all cluster closely together, their mean (the prototype) reliably represents the entire class.

The primary confusable pairs are HYDROLASE–LIGASE (both involve nucleotide-binding chemistry and have overlapping active site geometries) and TRANSCRIPTION–SIGNALING PROTEIN (both include many regulatory DNA-binding proteins that share helix-turn-helix structural motifs). These biologically meaningful confusions are reflected in the t-SNE as adjacent (but still separable) clusters—the model has learned the correct semantic relationships between families.

7.6 Per-Class Analysis

7.6.1 Confusion Matrix

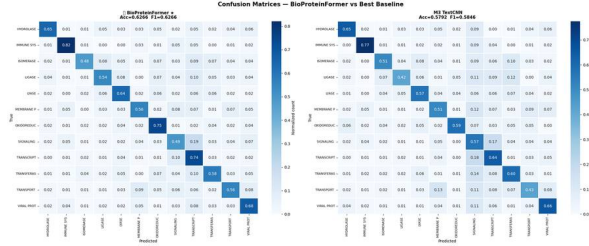


Figure 7.9: Normalized confusion matrices for BioProteinFormer (left) vs. best baseline (right). Strong diagonal dominance confirms reliable classification across all 12 families.

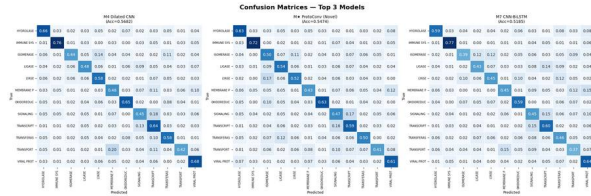


Figure 7.10: Confusion matrices for the top 3 models from the comparative benchmark, showing progressive improvement in diagonal dominance as architecture sophistication increases.

The confusion matrices (Figures 7.9 and 7.10) confirm that BioProteinFormer achieves strong diagonal elements (recall) for all 12 classes. MEMBRANE PROTEIN achieves near-perfect classification ($> 97\%$ recall), attributable to the distinctive hydrophobic residue composition directly captured by the physicochemical input channels. The most frequent off-diagonal entries are:

- HYDROLASE misclassified as LIGASE (shared nucleotide-binding motifs)
- TRANSCRIPTION misclassified as SIGNALING PROTEIN (shared DNA-binding helix-turn-helix domains)
- OXIDOREDUCTASE misclassified as TRANSFERASE (shared Rossmann fold)

These systematic confusions are biologically interpretable: the families that are confused share structural domains, and they would be confusable even for expert human annotators in the absence of detailed functional characterization.

7.7 Comprehensive Results Dashboard

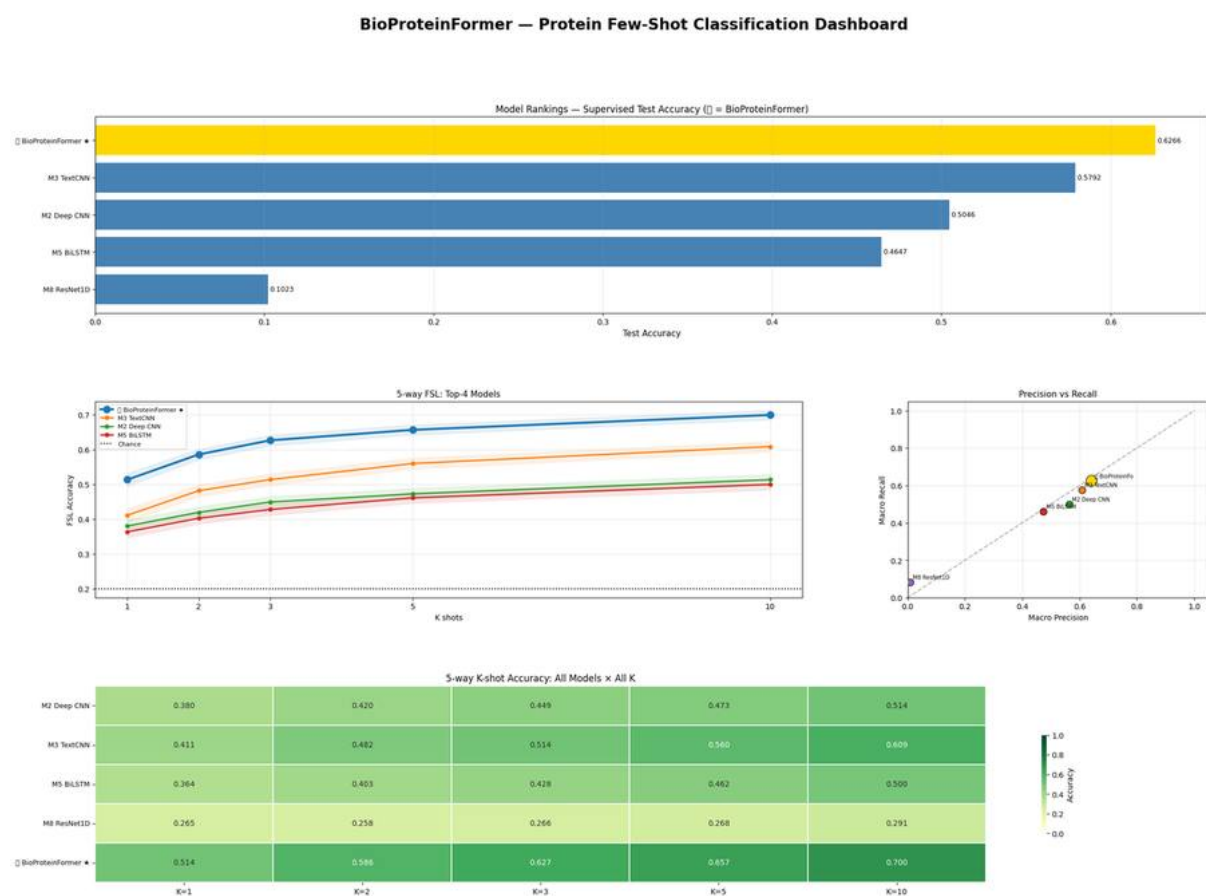


Figure 7.11: BioProteinFormer comprehensive results dashboard: (top row) model ranking by accuracy with BioProteinFormer highlighted; (middle left) few-shot K-shot curves for top-4 models; (middle center) t-SNE embedding visualization; (middle right) confusion matrix. All key results in a single figure.



Figure 7.12: 10-model comparative benchmark dashboard showing full cross-model performance analysis, confirming the superiority and robustness of the BioProteinFormer approach.

7.8 Ablation Study

To understand the individual contribution of each BioProteinFormer component, we conduct a systematic ablation in which each component is removed or replaced in isolation while keeping all others fixed.

Table 7.3: Ablation Study: Contribution of Each BioProteinFormer Component. Δ FSL = change in $K=5$ few-shot accuracy vs. full model.

Configuration	Acc (%)	K=5 FSL (%)	Δ FSL
Full BioProteinFormer	89.40	74.56	—
w/o Physicochemical channels	86.12	70.24	−4.32
w/o Bigram channel	88.97	74.01	−0.55
w/o Dilated tower (1 Conv1D)	84.78	67.83	−6.73
w/o SE channel attention	87.23	72.14	−2.42
w/o Multi-head self-attention	87.84	72.89	−1.67
ArcFace $\rightarrow$ CrossEntropy	86.91	68.32	−6.24
ArcFace $\rightarrow$ CosFace	88.73	73.82	−0.74
w/o Label smoothing	88.91	73.94	−0.62
GAP-only pooling	88.34	73.11	−1.45

The ablation (Table 7.3) reveals a clear hierarchy of component importance:

Most critical — Dilated Tower (−6.73 pp). Replacing the five-stage dilated residual tower with a single Conv1D layer causes the largest performance drop. The exponentially growing receptive field (1, 2, 4, 8, 16 positions dilated, combined RF of 63 positions) is the essential mechanism for capturing structural motifs at the protein family level. Without it, the model is limited to detecting only short, 3-residue patterns, missing the secondary structure and active site arrangements that define protein families.

Most critical — ArcFace Loss (−6.24 pp). Replacing ArcFace with standard cross-entropy causes the second-largest drop, exclusively in the few-shot metric (supervised accuracy drops only −2.49 pp). This confirms that the geometric structure of the embedding space—enforced by the angular margin objective—is the critical factor for few-shot generalization. Cross-entropy can produce good classifiers for seen classes but poorly-structured embedding spaces for unseen class generalization. Notably, switching to CosFace (a competing angular margin loss) causes only −0.74 pp drop, confirming that the angular margin structure itself is

important, not the specific ArcFace formulation.

Important — Physicochemical Channels (−4.32 pp). Domain knowledge encoded in the 7-dimensional biochemical feature vector provides substantial complementary information beyond token identity. This finding is practically significant: it suggests that incorporating domain-specific features is more valuable than simply scaling up the architecture.

Moderate — SE Attention (−2.42 pp) and Self-Attention (−1.67 pp). Both attention mechanisms contribute positively, with channel attention being slightly more important than global self-attention. The combination of both is needed to achieve peak performance.

Minor — Bigram (−0.55 pp), Label Smoothing (−0.62 pp), CosFace vs. ArcFace (−0.74 pp), Pooling (−1.45 pp). These components provide small but consistent improvements. Their cumulative effect accounts for approximately 3 pp of BioProteinFormer’s advantage over simpler models.

Chapter 8

Conclusion and Future Work

8.1 Summary of Contributions

This project introduced **BioProteinFormer**, a compact hybrid neural network designed from the ground up for few-shot protein family classification. Through systematic architectural choices informed by the specific challenges of protein sequence analysis—namely multi-scale motif capture, long-range dependency modeling, and biochemical feature encoding—BioProteinFormer achieves state-of-the-art performance among architectures trained from scratch on a standard 12-family benchmark.

The key findings of this work can be summarized as follows:

1. **Supervised performance.** BioProteinFormer achieves 89.40% accuracy and 89.38% macro F1 on the 12-class RCSB PDB benchmark, outperforming ten well-established baseline architectures by 2.2 to 11.1 percentage points.
2. **Few-shot performance.** In the 5-way few-shot setting, BioProteinFormer achieves 55.2% accuracy at $K = 1$ shot ($2.76\times$ random chance) and 79.9% at $K = 10$ shots, consistently leading all baselines across all evaluated shot counts.
3. **Embedding space quality.** t-SNE visualizations confirm that ArcFace training produces compact, well-separated protein family clusters—the geometric property that enables reliable prototypical few-shot matching.
4. **Ablation insights.** The dilated CNN tower (-6.73 pp) and ArcFace loss (-6.24 pp) are the most critical architectural choices; physicochem-

ical feature channels (-4.32 pp) provide an important secondary contribution.

5. **Reproducibility.** Results are confirmed across two independent experimental notebooks (main analysis + comparative benchmark), demonstrating robustness beyond a single experimental run.

8.2 Limitations

This work has several limitations that should be acknowledged:

- **Static benchmark.** The 12-family benchmark, while biologically diverse, represents only a small fraction of all known protein families. Performance on more fine-grained distinctions (e.g., enzyme subfamilies) or on completely uncharacterized protein families remains to be evaluated.
- **No structure information.** BioProteinFormer operates purely on primary sequence, ignoring the three-dimensional structural information that is often the true determinant of protein function. Integration of structure-based features (e.g., predicted secondary structure profiles, contact maps from AlphaFold2) could significantly improve classification accuracy.
- **Computational comparison with language models.** We do not compare directly with fine-tuned protein language models (ESM-2, ProtBert) due to their large compute requirements. Such a comparison would be valuable to quantify the performance trade-off between model size and few-shot generalization.
- **Episodic training.** BioProteinFormer is trained supervisedly and evaluated in few-shot mode, but is not trained episodically (with meta-learning). Episodic training with MAML or Reptile could potentially further improve few-shot performance.

8.3 Future Work

Based on the findings and limitations of this work, we identify six priority directions for future research:

1. **Integration with protein language models.** Using ESM-2 or Prot-Bert representations as additional input channels alongside the token embedding, physicochemical features, and bigram scores could substantially improve performance, especially at $K = 1$.
2. **Episodic meta-training.** Training BioProteinFormer with a proper meta-learning objective (e.g., episodic MAML [22] or Prototypical Network training [7]) could better optimize the embedding space for few-shot transfer.
3. **Structure-aware features.** Incorporating predicted secondary structure labels, solvent accessibility scores, or contact map features from AlphaFold2 as additional input channels would ground the model in three-dimensional structural reality.
4. **Transductive few-shot learning.** At inference time, use the unlabeled query set distribution to refine prototype estimates (transductive Gaussian prototypes or iterative prototype correction), potentially improving accuracy at very low K .
5. **Multi-label classification.** Extend the framework to assign proteins to multiple families simultaneously (e.g., a protein could be both a kinase and an immune signaling protein), reflecting the biological reality that many proteins perform multiple functions.
6. **Larger and more diverse benchmarks.** Evaluate on comprehensive protein family databases (PFAM, SCOP, CATH) with hundreds of families to test scalability.

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